SPARKY'S SCIENCE FACTS

What Are Simple Machines?

Engineering · Ages 7+

> **Short answer:** Simple machines are tools that make work easier, like levers, ramps, pulleys and wheels.

How it works

- A lever is a bar that turns on a point. A seesaw and a bottle opener are levers.
- A ramp, or inclined plane, lets you push something up slowly instead of lifting it straight up.
- A pulley is a wheel with a rope. It lets you pull down to lift something up, like a flag on a flagpole.
- A wheel and axle, like on a wheelbarrow, makes things easy to roll.
- Simple machines let you use a smaller push, but you have to push for longer.

Key words: Lever: a bar that turns on a pivot; Pivot: the point a lever turns on

Common mix-ups

- Machines do not make energy: they only change how you use your push.
- A longer lever makes lifting easier: push far from the pivot, close to the load.

Did you know? A wheelbarrow is a lever and a wheel together. It helps builders all over South Africa carry heavy bricks and sand.

Try it: Put a pencil under a ruler and a small book on one end. Press the other end to lift the book. Move the pencil and feel the difference!

Ask Sparky: Is a pair of scissors a machine? How does a crane lift heavy things?